

Solving Radical Equations and Checking Extraneous Roots

Answer Key

Algebra 2

Quick Answers

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|----------------|---------------|--------|-----------------|
| 1. 28 | 4. 5, 0, 5, 2 | 7. , 6 | 10. 4, -1, 4, 4 |
| 2. 8 | 5. 100 | 8. 4 | |
| 3. -2, 3, 0, 4 | 6. 150 | 9. 9 | |
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Curriculum

Level: Algebra 2

HSA-REI.B.4 – problems 1, 3, 4, 5, 6, 7, 9, 10

HSA-SSE.B.3 – problems 2, 8

Difficulty 1–5 of 5 across 17 tagged checks.

1. Solve $\sqrt{x-3} = 5$.

Solution: the radical is already isolated, so square both sides: $x - 3 = 25$, hence $x = 28$. Check: $\sqrt{28-3} = \sqrt{25} = 5$ ✓. Both sides were non-negative before squaring, so no extraneous root could appear. $\boxed{x = 28}$

2. Rewrite and evaluate $16^{3/4}$.

Solution: a rational exponent is root-then-power: $16^{3/4} = (\sqrt[4]{16})^3 = 2^3 = 8$. Taking the fourth root first keeps the arithmetic small; the other order needs $16^3 = 4096$ before rooting. $\boxed{8}$

3. $\sqrt{x+6} = x$, squared equation $x^2 - x - 6 = 0$.

(a) $x^2 - x - 6 = (x-3)(x+2) = 0$ gives the candidates $\boxed{x = -2}$ and $\boxed{x = 3}$.

(b) Substituting into $\sqrt{x+6} = x$: at $x = 3$: $\sqrt{9} - 3 = 0$ ✓; at $x = -2$: $\sqrt{4} - (-2) = 2 + 2 = \boxed{4} \neq 0$ ✗.

(c) $x = -2$ is **extraneous** — it solves $x+6 = x^2$ but makes the original read $2 = -2$, and the principal square root is never negative. Solution: $\boxed{x = 3}$

4. Solve $\sqrt{3x+1} = x-1$.

(a) Squaring: $3x+1 = x^2 - 2x + 1$, so $x^2 - 5x = 0$ and $x(x-5) = 0$, giving candidates $\boxed{x = 0}$ and $x = 5$.

(b) At $x = 0$: $\sqrt{3(0)+1} - (0-1) = 1 + 1 = \boxed{2} \neq 0$ ✗ — the left side is 1 while the right side is -1, so this candidate is extraneous. At $x = 5$: $\sqrt{16} = 4$ and $5-1 = 4$ ✓.

(c) Solution set: $\boxed{x = 5}$

Why it happened: the right side $x-1$ is negative at $x = 0$, and a square root can never equal a negative number.

5. Free fall: $t = \sqrt{h/16}$ with $t = 2.5$ seconds; find h in feet.

Solution: substitute and solve the radical equation:

$$2.5 = \sqrt{\frac{h}{16}} \implies 6.25 = \frac{h}{16} \implies h = 100$$

Check in the model: $\sqrt{100/16} = \sqrt{6.25} = 2.5 \checkmark$. Both sides are non-negative here (a time and a principal root), so the candidate cannot be extraneous. $\boxed{h = 100 \text{ ft}}$

6. Skid marks: $s = \sqrt{24x}$ with $s = 60$ miles per hour; find x in feet.

Solution:

$$60 = \sqrt{24x} \implies 3600 = 24x \implies x = 150$$

Check: $\sqrt{24 \cdot 150} = \sqrt{3600} = 60 \checkmark$. $\boxed{x = 150 \text{ ft}}$

Domain note: only $x \geq 0$ makes sense as a skid length, and the model's own square root already refuses negative inputs.

7. Solve $\sqrt{x-2} = -3$, or explain why there is no real solution.

Solution: squaring gives $x - 2 = 9$, so $x = 11$ is the only candidate. Substituting into $\sqrt{x-2} + 3$ at $x = 11$: $\sqrt{9} + 3 = \boxed{6} \neq 0$, i.e. $\sqrt{9} = 3 \neq -3 \times$.

The principal square root outputs only values ≥ 0 , so $\sqrt{x-2} = -3$ is impossible for every real x ; the candidate is extraneous and the equation has $\boxed{\text{no real solution}}$

This is the cleanest illustration of why the check is mandatory: squaring manufactured a root out of an equation with none.

8. Solve $x^{3/2} = 8$.

Solution: in radical form the equation is $(\sqrt{x})^3 = 8$. Take the cube root of both sides: $\sqrt{x} = 2$, then square: $x = 4$. Check: $4^{3/2} = (\sqrt{4})^3 = 2^3 = 8 \checkmark$. $\boxed{x = 4}$

No negative solution: $x^{3/2}$ contains $\sqrt{x}$, which is undefined for $x < 0$ in the real numbers — the domain rules negatives out before any solving happens.

9. Solve $\sqrt{x+7} - \sqrt{x} = 1$.

Solution: isolate one radical first, or squaring leaves two of them:

$$\begin{aligned} \sqrt{x+7} &= 1 + \sqrt{x} \\ x + 7 &= 1 + 2\sqrt{x} + x && \text{(square both sides)} \\ 6 &= 2\sqrt{x} \implies \sqrt{x} = 3 && \text{(isolate the radical)} \\ x &= 9 && \text{(square again)} \end{aligned}$$

Check: $\sqrt{9+7} - \sqrt{9} = 4 - 3 = 1 \checkmark$. $\boxed{x = 9}$

10. $\sqrt{x+5} = x-1$ with the table of both sides.

(a) At $x = -1$: $\sqrt{4} = 2$ but $x - 1 = -2$ — *not equal*. At $x = 4$: $\sqrt{9} = 3$ and $x - 1 = 3$ — *equal*. (Squaring gives $x + 5 = x^2 - 2x + 1$, i.e. $x^2 - 3x - 4 = 0$, whose roots are the two candidates $x = -1$ and $x = 4$.)

(b) Solution set: $x = 4$

(c) The curve $y = \sqrt{x+5}$ lies on or above the x -axis everywhere, so it can only meet $y = x - 1$ where that line is non-negative, i.e. for $x \geq 1$ — one intersection. Squaring replaces the line by the parabola $y = (x - 1)^2$, which is non-negative on both sides of $x = 1$, so it also meets the graph of $y = x + 5$ at $x = -1$. The table shows the cost: at $x = -1$ the two original sides differ by 4 , so that intersection belongs to the squared equation only.